



**Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 8 • STANDARD 6.AT.8

# Equations and Inequalities Problem Solving

Lesson 8-7 • Teacher Copy

**Name:** \_\_\_\_\_

**Date:** \_\_\_\_\_

**Class:** \_\_\_\_\_

## My Notes

Level 1 Support



### TODAY'S OBJECTIVES

**Content Objective:** I can model and solve real-world problems using equations and inequalities.

**Language Objective:** I can explain my reasoning using the words model, equation, inequality, and reasonableness.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



**Fill in each blank as we go. Use the Word Bank to help you.**



### WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Equations and Inequalities Problem Solving

Model

Equation

Inequality

Reasonableness

Variable

Constraint



Tap any word to see what it means and a picture.

1

— Use an equation for one exact value or an inequality for a range of values in a real situation.

2

A drawing, equation, or diagram that represents a situation is a

.

3

A math sentence with an equal sign is an .

4

A math sentence that compares values with  $<$ ,  $>$ ,  $\leq$ , or  $\geq$  is an

.

5

Checking whether an answer makes sense for the problem is checking its

.

- 6 A letter that stands for the unknown amount in a problem is a

.

- 7 A limit that tells which values are allowed in a problem is a

.

## Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

### 1. Understand the Question

EXPLAIN

**How much more can the detective spend? How much does each flashlight cost? When do you use an equation versus an inequality?**

**Your job:** Answer the question. Use math evidence. Explain how the evidence proves your answer.

*Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.*

## 2. Plan Your Math Words

Check at least two words you will use.

☐

### **Equations and Inequalities**

**Problem Solving** — Use an equation for one exact value or an inequality for a range of values in a real situation.

*Resolución de problemas con ecuaciones y desigualdades — Usar una ecuación para un valor exacto o una desigualdad para un rango de valores en una situación real.*

☐

**Model** — To show a real-life situation with an equation or inequality.

*Modelar — Mostrar una situación real con una ecuación o desigualdad.*

☐

**Equation** — A math sentence with an equal sign showing both sides are the same.

*Ecuación — Una oración matemática con un signo igual que muestra que ambos lados son iguales.*

☐

**Inequality** — A math sentence that compares two sides with  $<$ ,  $>$ ,  $\leq$ , or  $\geq$ .

*Desigualdad — Una oración matemática que compara dos lados con  $<$ ,  $>$ ,  $\leq$  o  $\geq$ .*

☐

**Reasonableness** — Checking if your answer makes sense.

*Razonabilidad — Revisar si tu respuesta tiene sentido.*

**Say it first:** Say your idea to a partner or quietly to yourself before you write.

*Di tu idea a un compañero o en voz baja antes de escribir.*

**The detective can spend at most \$\_\_\_ more because  $r \leq 500 - \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$ .**

**Each flashlight costs \$\_\_\_ because  $f = 60 \div \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$ .**

*Di tu respuesta primero: Mi respuesta es \_\_\_ porque \_\_\_.*

### 3. Build Your Explanation

**Model the parts:** This model shows the parts of an explanation. It does not answer your writing question.

**Claim:** An exact total points to an equation; a limit or range points to an inequality.

**Evidence:** Students use signal words ('equals/total' for an equation; 'at least/at most/no more than' for an inequality) to choose the correct model.

**Reasoning:** This evidence connects the definition of equations and inequalities problem solving to the mathematical claim.

#### Start

Most language support

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- **The detective can spend at most \$\_\_ more because  $r \leq 500 - \_\_ = \_\_$ . Each flashlight costs \$\_\_ because  $f = 60 \div \_\_ = \_\_$ .**
- **My answer is \_\_.**

*Mi respuesta es \_\_.*

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#### Build

Some language support

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- **My claim is \_\_.**
- **I know because \_\_.**

*Mi afirmación es \_\_.*

*Lo sé porque \_\_.*

- **So \_\_.**

*Entonces \_\_.*

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## Explain

### Light language support

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- **My claim is \_\_\_\_.**  
*Mi afirmación es \_\_\_\_.*
- **My math evidence is \_\_\_\_.**  
*Mi evidencia matemática es \_\_\_\_.*
- **This proves \_\_\_\_ because \_\_\_\_.**  
*Esto demuestra \_\_\_\_ porque \_\_\_\_.*

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## 4. Check Your Explanation

- ☐ I answered the question.  
*Respondí la pregunta.*
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.  
*Usé evidencia matemática: números, una ecuación, un modelo o una comparación.*
- ☐ I explained how the evidence proves my answer.  
*Explicué cómo la evidencia demuestra mi respuesta.*
- ☐ I used at least two math words.  
*Usé por lo menos dos palabras matemáticas.*
- ☐ I reread complete sentences and punctuation.  
*Volví a leer mis oraciones completas y la puntuación.*

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## Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

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## Answer Key & Teacher Guide

1. **Notes 1:** Equations and Inequalities Problem Solving
2. **Notes 2:** Model
3. **Notes 3:** Equation
4. **Notes 4:** Inequality
5. **Notes 5:** Reasonableness
6. **Notes 6:** Variable
7. **Notes 7:** Constraint
8. **Watch for this mistake:** *A common mistake in Equations and Inequalities Problem Solving is matching the wrong model to a key phrase. For 'the getaway car held no more than 6 bags,' students often write the equation  $b = 6$  instead of the inequality  $b \leq 6$ , because they see one number and assume it must be exact. The same mistake happens in reverse: for 'the reward split equally among 5 people gave each \$40,' students sometimes write the inequality  $5p \geq 40$  instead of the equation  $5p = 40$ , because 'split equally' sounds like it could mean a range. Before you submit, ask: does this phrase give one exact value (equation,  $=$ ) or a limit/range (inequality,  $<$ ,  $>$ ,  $\leq$ ,  $\geq$ )?*
9. **Try It 1:** A.  $x \leq 20$  — 'At most 20' means 20 or fewer, so  $x \leq 20$ .
10. **Try It 2:** A.  $w = 12$  — Divide both sides by 5:  $w = 60 \div 5 = 12$  pounds. Check:  $5 \times 12 = 60$  ✓